

SHAULI RAVFOGEL

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RESEARCH INTERESTS

Interpretability and control of generative neural models, particularly language models. My work develops mathematically-principled methods for analyzing and intervening on the internal representations of LMs, with emphasis on concept erasure, counterfactual generation, causal attribution, and the emergence of structured (often linear) representations as a byproduct of the language-modeling objective.

ACADEMIC POSITIONS

- **2025–present:** Faculty Fellow, Center for Data Science, New York University.
- **Sep–Dec 2024:** Visiting Researcher, ETH Zürich (Prof. Ryan Cotterell’s group).
- **Oct 2023–Sep 2024:** Student Researcher, Google Research.
- **Jun–Sep 2023:** Research Intern, Bloomberg, London.
- **Jun–Sep 2022:** Research Intern, Facebook AI Research, Israel.
- **Jan 2020–2023:** PhD Research Intern, Allen Institute for AI (AI2), Israel.

EDUCATION

- **2020–2024:** PhD in Computer Science, Bar-Ilan University.
Dissertation: *Analyzing the Representation Space of Transformer-Based Language Models* (IAAI Best PhD Thesis Award).
Advisor: Prof. Yoav Goldberg.
- **2018–2020:** MSc in Computer Science, Bar-Ilan University. Advisor: Prof. Yoav Goldberg.
- **2015–2018:** BSc in Computer Science, Bar-Ilan University.
- **2010–2013:** BSc in Chemistry, Bar-Ilan University.

AWARDS AND FELLOWSHIPS

- Israeli Council for Higher Education (Vatat) Postdoctoral Fellowship in Data Science (2025).
- IAAI Best PhD Thesis Award (2024).
- Blavatnik Prize for Outstanding Israeli Doctoral Students in Computer Science (2024).
- Fulbright Postdoctoral Fellowship (2024; declined for the NYU CDS Faculty Fellowship).
- DAAD AInet Postdoctoral Fellowship (2023).
- Outstanding Paper Award, Bar-Ilan Data Science Institute (2023).
- Bloomberg Data Science PhD Fellowship (2022–2024; first recipient outside the US).
- Israeli Council for Higher Education (Vatat) Fellowship for Outstanding PhD Students in Data Science (2022; declined due to a conflict).
- Bar-Ilan Presidential Scholarship for Outstanding PhD Students (2021–2024).
- Best Paper Award, RepL4NLP Workshop, ACL 2022.

GRANTS AND FUNDING

- **Co-Principal Investigator**, *LLM Introspection*. Arena Academic Partnerships Program (with Tal Linzen). NYU Center for Data Science, 2026.

PUBLICATIONS

* *denotes equal contribution.*

Refereed Publications — First or Last Author

- **Shauli Ravfogel**, Gilad Yehudai, Tal Linzen, Joan Bruna, and Alberto Bietti. Emergence of Linear Truth Encodings in Language Models. NeurIPS 2025.
- Floris Holstege*, **Shauli Ravfogel***, and Bram Wouters. Preserving Task-Relevant Information Under Linear Concept Removal. NeurIPS 2025.
- **Shauli Ravfogel***, Anej Svete*, Vésteinn Snæbjarnarson, and Ryan Cotterell. Gumbel Counterfactual Generation from Language Models. ICLR 2025.
- Matan Avitan, Ryan Cotterell, Yoav Goldberg, and **Shauli Ravfogel**. Natural Language Counterfactuals through Representation Surgery. Findings of NAACL 2025.
- **Shauli Ravfogel**, Valentina Pyatkin, Amir David Nissan Cohen, Avshalom Manevich, and Yoav Goldberg. Description-based Text Similarity. COLM 2024.
- Shashwat Singh*, **Shauli Ravfogel***, Jonathan Herzig, Roei Aharoni, Ryan Cotterell, and Ponurangam Kumaraguru. Representation Surgery: Theory and Practice of Affine Steering. ICML 2024.
- Aviv Slobodkin, Omer Goldman, Avi Caciularu, Ido Dagan, and **Shauli Ravfogel**. The Curious Case of Hallucinatory Unanswerability: Finding Truths in the Hidden States of Over-Confident Large Language Models. EMNLP 2023.
- **Shauli Ravfogel**, Yoav Goldberg, and Ryan Cotterell. Log-linear Guardedness and its Implications. ACL 2023.
- **Shauli Ravfogel**, Yoav Goldberg, and Jacob Goldberger. Conformal Nucleus Sampling. Findings of ACL 2023.
- **Shauli Ravfogel**, Francisco Vargas, Yoav Goldberg, and Ryan Cotterell. Kernelized Concept Erasure. EMNLP 2022.
- Royi Rassin, **Shauli Ravfogel***, and Yoav Goldberg. DALL-E 2 is Seeing Double: Flaws in Word-to-Concept Mapping in Text2Image Models. BlackboxNLP Workshop, EMNLP 2022.
- **Shauli Ravfogel**, Michael Twiton, Yoav Goldberg, and Ryan D. Cotterell. Linear Adversarial Concept Erasure. ICML 2022.
- **Shauli Ravfogel***, Grusha Prasad*, Tal Linzen, and Yoav Goldberg. Counterfactual Interventions Reveal the Causal Effect of Relative Clause Representations on Agreement Prediction. CoNLL 2021.
- Carlo Meloni*, **Shauli Ravfogel***, and Yoav Goldberg. Ab Antiquo: Neural Proto-language Reconstruction. NAACL 2021.
- **Shauli Ravfogel***, Yanai Elazar*, Jacob Goldberger, and Yoav Goldberg. Unsupervised Distillation of Syntactic Information from Contextualized Word Representations. BlackboxNLP Workshop, EMNLP 2020.

- **Shauli Ravfogel**, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection. ACL 2020.
- **Shauli Ravfogel**, Yoav Goldberg, and Tal Linzen. Studying the Inductive Biases of RNNs with Synthetic Variations of Natural Languages. NAACL 2019.
- **Shauli Ravfogel**, Yoav Goldberg, and Francis Tyers. Can LSTM Learn to Capture Agreement? The Case of Basque. BlackboxNLP Workshop, EMNLP 2018.

Refereed Publications — Co-author

- Or Shafran, Shaked Ronen, Omri Fahn, **Shauli Ravfogel**, Atticus Geiger, and Mor Geva. From Directions to Regions: Decomposing Activations in Language Models via Local Geometry. ICML 2026.
- Yu Fan, Yang Tian, **Shauli Ravfogel**, Mrinmaya Sachan, Elliott Ash, and Alexander Miserlis Hoyle. The Medium Is Not the Message: Deconfounding Document Embeddings via Linear Concept Erasure. EMNLP 2025.
- Yihuai Hong, Lei Yu, **Shauli Ravfogel**, Haiqin Yang, and Mor Geva. Intrinsic Evaluation of Unlearning Using Parametric Knowledge Traces. EMNLP 2025.
- Zhiqi Huang, Puxuan Yu, **Shauli Ravfogel**, and James Allan. Language Concept Erasure for Language-invariant Dense Retrieval. EMNLP 2024.
- Royi Rassin, Eran Hirsch, Daniel Glickman, **Shauli Ravfogel**, Yoav Goldberg, and Gal Chechik. Linguistic Binding in Diffusion Models: Enhancing Attribute Correspondence through Attention Map Alignment. NeurIPS 2023.
- Nora Belrose, David Schneider-Joseph, **Shauli Ravfogel**, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect Linear Concept Erasure in Closed Form. NeurIPS 2023.
- Mosh Levy, **Shauli Ravfogel**, and Yoav Goldberg. Guiding LLM to Fool Itself: Automatically Imbuing Multiple Shortcut Triggers in Question Answering Samples. Findings of EMNLP 2023.
- Marius Mosbach, Tiago Pimentel, **Shauli Ravfogel**, Dietrich Klakow, and Yanai Elazar. Few-shot Fine-tuning vs. In-context Learning: A Fair Comparison and Evaluation. Findings of ACL 2023.
- Rita Sevastjanova, Eren Cakmak, **Shauli Ravfogel**, Ryan Cotterell, and Mennatallah El-Assady. Visual Comparison of Language Model Adaptation. IEEE Transactions on Visualization and Computer Graphics 2022.
- Elad Ben-Zaken, **Shauli Ravfogel**, and Yoav Goldberg. BitFit: Simple Parameter-efficient Fine-tuning for Transformer-based Masked Language-models. ACL 2022.
- Hila Gonen, **Shauli Ravfogel**, and Yoav Goldberg. Analyzing Gender Representation in Multilingual Models. RepL4NLP Workshop, ACL 2022 (**Best Paper Award**).
- Yanai Elazar, Nora Kassner, **Shauli Ravfogel**, Abhilasha Ravichander, Eduard Hovy, Hinrich Schütze, and Yoav Goldberg. Measuring and Improving Consistency in Pretrained Language Models. Transactions of the Association for Computational Linguistics (TACL) 2021.
- Yanai Elazar, **Shauli Ravfogel**, Alon Jacovi, and Yoav Goldberg. Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals. Transactions of the Association for Computational Linguistics (TACL) 2021.
- Alon Jacovi, Swabha Swayamdipta, **Shauli Ravfogel**, Yanai Elazar, Yejin Choi, and Yoav Goldberg. Contrastive Explanations for Model Interpretability. EMNLP 2021.

- Hila Gonen, **Shauli Ravfogel**, Yanai Elazar, and Yoav Goldberg. It's Not Greek to mBERT: Inducing Word-Level Translations from Multilingual BERT. BlackboxNLP Workshop, EMNLP 2020.

Preprints and Manuscripts Under Review

- Shashwat Singh, Tal Linzen, **Shauli Ravfogel**. Can LLMs Introspect? A Reality Check. Under review.
- Zhuofan Josh Ying, **Shauli Ravfogel**, Nikolaus Kriegeskorte, and Peter Hase. The Truthfulness Spectrum Hypothesis. Preprint, 2026.
- **Shauli Ravfogel**, Gilad Yehudai, Tal Linzen, Joan Bruna, and Alberto Bietti. Geometric Factual Recall in Transformers. Under review.
- Mosh Levy, Zohar Elyoseph, **Shauli Ravfogel**, and Yoav Goldberg. State over Tokens: Characterizing the Role of Reasoning Tokens. Under review.
- Jackson Petty, Michael Y. Hu, Wentao Wang, **Shauli Ravfogel**, William Merrill, and Tal Linzen. RELIC: Evaluating Compositional Instruction Following via Language Recognition. Under review.
- Royi Rassin, Aviv Slobodkin, **Shauli Ravfogel**, Yanai Elazar, and Yoav Goldberg. GRADE: Quantifying Sample Diversity in Text-to-Image Models. Under review.

TEACHING

- **Lecturer:** *Causality and Interpretability of Language Models*, New York University (Spring 2025).
- **Lecturer:** *Introduction to Machine Learning*, Bar-Ilan University (2020–2021).
- **Teaching Assistant:** Graduate Machine Learning, Bar-Ilan University (2019–2021).

MENTORING

Co-advised several master's students (further details available upon request).

INVITED TALKS

- Penn NLP Group, University of Pennsylvania (Spring 2026).
- Aspen Meeting on Foundation Models (Autumn 2025).
- UK AI Safety Institute (AIS) Alignment Conference (Autumn 2025).
- Cambridge Language Technology Lab, University of Cambridge (Autumn 2025).
- Technion — Israel Institute of Technology (Winter 2024).
- EPFL — École polytechnique fédérale de Lausanne (Autumn 2023).
- DeepMind (Summer 2023).
- J. P. Morgan, London (Summer 2023).
- IBM Research, Israel (2023).
- Hebrew University NLP Group (2023).
- SIGTYP Lecture Series (2021).
- Prof. Roi Reichart's group, Technion (2020).
- Prof. Robert Frank's group, Yale University (2020).

ACADEMIC SERVICE

- **Senior Area Chair:** EMNLP.
- **Area Chair:** ACL Rolling Review (ARR), COLM.
- **Reviewer:** ACL, EMNLP, NAACL, NeurIPS, ICML.
- **Grant Reviewer:** U.S.–Israel Binational Science Foundation (BSF), Schmidt Science.
- **Workshop Co-organizer:** Representation Learning for NLP (RepL4NLP), ACL 2023 (250+ attendees).

RESEARCH VISITS

- 2021–2023: Prof. Ryan Cotterell’s group, ETH Zürich (three visits).
- 2020: Prof. Tal Linzen’s group, Johns Hopkins University.